Tor Vergata University of Rome

FACULTY OF ENGINEERING

COMPUTER SCIENCE, CONTROL AND GEOINFORMATION
CYCLE XXXVII

PhD Thesis

TITLE

ADVISOR

Prof. Sergio Galeani

CANDIDATE

Roberto Masocco

COORDINATOR

Prof. Francesco Quaglia

A.Y. 2023/2024

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Wings are a constraint that makes
it possible to fly.
— Robert Bringhurst

To my parents…

Acknowledgements

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Lausanne, 12 Mars 2011 D. K.

Preface

A preface is not mandatory. It would typically be written by some other person (eg your thesis director).

Footnote should be cited here Galeani, Masocco, and Sassano 2022; Galperti et al. 2024.

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Lausanne, 12 Mars 2011 T. D.

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1 Introduction

In the history of mankind, many turning points have been marked by technological advancements. From the invention of the wheel to the steam engine, from the light bulb to the computer, technology has always changed the way we live, work, and see the world. Its impact on society has always been profound and sometimes positive, sometimes negative. The changes it brought had consequences on our daily habits, introducing new ways of doing things, and sometimes even new ways of thinking. With the same spirit that drove our ancestors to develop *tools* made of stones and sticks to make their lives easier while harvesting food, we are now developing ever more powerful machines of all sorts to handle the challenges of the modern world, either in conjunction with a human operator or in complete autonomy. Moreover, considering the recent developments of artificial intelligence, these *tools* have reached such a level of sophistication that they may also reshape our thought process in the near future.

The technological advancements that brought us these new solutions increased not only their capabilities but also their complexity and, ultimately, their variety. A good example of this is the industrial manipulator robot: a machine that can perform a wide range of tasks, from simple pick-and-place operations to more complex ones, such as welding, painting, or even surgery. These machines have evolved from mechanical arms composed by multiple actuators, all connected to a single power and computational unit, to a network of sensors and microcontrollers, grouped in subsystems all interconnected to a central unit, which is in turn responsible for their coordination, supervision, and control, as well as for interfacing with human operators. This evolution has been driven, in this particular case, by the need to increase the robot's capabilities, making it more versatile and adaptable to different tasks, and by the need to make it more reliable and safe, reducing the risk of accidents and the downtime due to maintenance. Another suitable example would be the airplane: considering even a small civilian aircraft, originally equipped with an engine, control surfaces and a few instruments, it has evolved into a complex system of systems, composed by a multitude of sensors, probes, indicators and avionics. In this scenario, too, the evolution has been driven by specific needs still related to safety and comfort, but ultimately summarisable in the need to ease the lives of both the passengers and the crew.

All of this innovation comes at a price: when the multitude of subsystems, subcomponents, and submodules increases, new problems arise. The complexity of the system grows, and so does the difficulty of understanding it, controlling it, and maintaining it. The interactions between the different parts of the system become more intricate, and the consequences of a failure in one of them can propagate to the others, causing a cascade of failures that can lead to catastrophic events. A deep understanding is needed to prevent events like these, for which a good organization of the system *architecture* is of paramount importance. Moreover, a well-organized architecture may also be incredibly beneficial in two other stages: the design and development of the system, and its maintenance. In the former, a clear and well-structured architecture naturally leads to a clear and well-structured design, following a top-down approach that starts from the system's requirements and ends with the definition of the subsystems and their interfaces, each following a common set of standards and best practices that adhere to the structure of the architecture itself. In the latter, a well-structured architecture allows for a clear identification of the subsystems and their interactions, easing the task of diagnosing and fixing a failure or an error. Finally, a good architecture can also be a powerful resource for the evolution of the system as a whole, allowing for the addition of new functionalities or the replacement of old components with new ones, without the need to redesign the whole system from scratch, greatly reducing the time and the costs of such an operation.

This work is focused on software architectures for distributed systems for control and autonomy.

Just like our ancestors that built the first tools, we follow the same approach: identify the problem, lay down the requirements, design the solution, build it, test it, and finally use it.

A An appendix

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